

Ipopt revisited in Rust

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Agenda

- A tour of Ipopt

 - History

 - Techniques applied

- An attempt of a simple IPM algorithm

 - Challenges encountered

 - Results

- A tour of ripopt

 - Context

 - Benchmarks

- Conclusion

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Ipopt: Interior Point OPTimizer

Popular open-source non-linear optimization package

Bindings available for:

- ▶ AMPL (algebraic modeling language)
- ▶ Fortran
- ▶ GAMS (general algebraic modeling system)
- ▶ Julia
- ▶ Matlab / Scilab
- ▶ .NET languages: C#, F# and Visual Basic.NET
- ▶ Python (several wrappers available)
- ▶ R
- ▶ Rust

Ipopt: The origins

- ▶ PhD thesis by Andreas Wächter in 2002 at Carnegie Mellon¹
- ▶ Since 2002 part of the COIN-OP foundation
- ▶ Original implementation in Fortran 77 (1.x and 2.x), now deprecated
- ▶ Open source since the start
- ▶ Benchmarked against KNITRO (1999)² and LOQO (1999)³

¹Wachter 2002.

²Byrd, Hribar, and Nocedal 1999.

³Vanderbei 1999.

Ipopt: Subsequent paper

- ▶ Paper by Wächter and Biegler in 2006⁴
- ▶ More focused on the central algorithm of Ipopt
- ▶ Features described more in detail
 - ▶ Feasibility restoration phase,
 - ▶ Inertia correction,
 - ▶ Various heuristics...
- ▶ More benchmarks: CUTEr test set⁵

⁴Wächter and Biegler 2006.

⁵Gould, Orban, and Toint 2003.

Ipopt: C++ version

- ▶ IBM Research decided to invest in an open source re-write of Ipopt in C++
- ▶ With the help of Carl Laird, the code was re-implemented from scratch ⁶
- ▶ 3.x published in Aug 2005
- ▶ It is still actively developed on GitHub

⁶History of Ipopt

Ipopt: Summary of applied techniques

- ▶ Primal-dual interior-point method with filter line-search globalization
- ▶ Sparse symmetric KKT solves with inertia monitoring/correction
- ▶ Inertia correction and regularization to obtain the proper search direction
- ▶ Feasibility restoration phase for difficult or infeasible iterates
- ▶ Second-order correction steps to reduce Maratos-type effects
- ▶ Practical heuristics for robustness and faster performance

Interior Point Methods: Intro

$$\min_{x \in \mathbb{R}^n} \quad \varphi_\mu(x) := f(x) - \mu \sum_{i=1}^n \ln(x^{(i)}) \quad (1)$$

$$\text{s.t.} \quad c(x) = 0 \quad (2)$$

$$x \geq 0 \quad (3)$$

We recover the Karush-Kuhn-Tucker (KKT) conditions for $\mu = 0$:

$$\nabla f(x) + \nabla c(x)\lambda - z = 0 \quad (4)$$

$$c(x) = 0 \quad (5)$$

$$XZe - \mu e = 0 \quad (6)$$

$$x, z \geq 0, \quad \lambda \in \mathbb{R}^m, \quad z \in \mathbb{R}^n,$$

$$Z := \text{diag}(z), e = (1, \dots, 1)^T$$

n : number of primal variables, m : number of equality constraints

KKT system: Derivation

$$F_{\mu}(x, \lambda, z) = \begin{bmatrix} \nabla f(x) + \nabla c(x)\lambda - z \\ c(x) \\ XZe - \mu e \end{bmatrix} = 0$$

Iteratively solve using Newton's method:

- ▶ Fix a starting point (x_k, λ_k, z_k) ,
- ▶ Calculate the deltas for each dimension $(d_k^x, d_k^{\lambda}, d_k^z)$
- ▶ Use the deltas to move to $(x_{k+1}, \lambda_{k+1}, z_{k+1})$ (more later)

$$F_{\mu_j}(x_k, \lambda_k, z_k) + \nabla F_{\mu_j}(x_k, \lambda_k, z_k)(d_k^x, d_k^{\lambda}, d_k^z)^T = 0$$

The term $\nabla F_{\mu}(x, \lambda, z)$ is a 3x3 matrix.

It is the Jacobian of the vector field.

We can move $F_{\mu}(x, \lambda, z)$ to the other side to get the form $Ax = b$.

KKT system: Matrix form

$$\begin{bmatrix} W_k & A_k & -I \\ A_k^T & 0 & 0 \\ Z_k & 0 & X_k \end{bmatrix} \begin{bmatrix} d_k^x \\ d_k^\lambda \\ d_k^z \end{bmatrix} = - \begin{bmatrix} \nabla f(x_k) + A_k \lambda_k - z_k \\ c(x_k) \\ X_k Z_k e - \mu_j e \end{bmatrix}$$

$$\text{where } A_k := \nabla c(x_k),$$

$$W_k := \nabla_{xx}^2 \mathcal{L}(x_k, \lambda_k, z_k),$$

$$\mathcal{L}(x, \lambda, z) := f(x) + c(x)^T \lambda - z.$$

But this system can be further simplified!

Multiply out the third equation:

$$Z_k d_k^x + X_k d_k^z = \mu_j e - X_k Z_k e$$

$$d_k^z = \mu_j X_k^{-1} e - z_k - X_k^{-1} Z_k d_k^x$$

$$\Sigma_k := X_k^{-1} Z_k$$

KKT system: Symmetric matrix form

Now d_k^z only appears in the first equation as $-ld_k^z = -d_k^z$ which we move to the other side:

$$\begin{aligned}W_k d_k^x + A_k d_k^\lambda &= -(\nabla f(x_k) + A_k \lambda_k - z_k) + d_k^z \\W_k d_k^x + A_k d_k^\lambda &= -(\nabla f(x_k) + A_k \lambda_k - z_k) + \\&\quad + \mu_j X_k^{-1} e - z_k - \Sigma_k d_k^x \\(W_k + \Sigma_k) d_k^x + A_k d_k^\lambda &= -(\nabla f(x_k) + A_k \lambda_k - \mu_j X_k^{-1} e) \\(W_k + \Sigma_k) d_k^x + A_k d_k^\lambda &= -(\nabla \varphi_{\mu_j}(x_k) + A_k \lambda_k)\end{aligned}$$

Reconstructing the system, we get:

$$\begin{bmatrix} W_k + \Sigma_k & A_k \\ A_k^T & 0 \end{bmatrix} \begin{bmatrix} d_k^x \\ d_k^\lambda \end{bmatrix} = - \begin{bmatrix} \nabla \varphi_{\mu_j}(x_k) + A_k \lambda_k \\ c(x_k) \end{bmatrix} \quad (7)$$

KKT system: Important considerations

$$\begin{bmatrix} W_k + \Sigma_k & A_k \\ A_k^T & 0 \end{bmatrix} \begin{bmatrix} d_k^x \\ d_k^\lambda \end{bmatrix} = - \begin{bmatrix} \nabla \varphi_{\mu_j}(x_k) + A_k \lambda_k \\ c(x_k) \end{bmatrix}$$

- ▶ Symmetric system
- ▶ Matrix in top-left block projected onto the null space of the constraint Jacobian A_k^T is positive definite
- ▶ If A_k does not have full rank, the iteration matrix is singular and a solution may not exist

Solve instead the following system with **inertia correction** for some $\delta_w, \delta_c \geq 0$:

$$\begin{bmatrix} W_k + \Sigma_k + \delta_w I & A_k \\ A_k^T & -\delta_c I \end{bmatrix} \begin{bmatrix} d_k^x \\ d_k^\lambda \end{bmatrix} = - \begin{bmatrix} \nabla \varphi_{\mu_j}(x_k) + A_k \lambda_k \\ c(x_k) \end{bmatrix}$$

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Introduction

I wanted to play around with code and not just present equations
and thus produce some nice visualizations :)

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I wanted to play around with code and not just present equations
and thus produce some nice visualizations :)

I wanted to see if optimization in pure Rust is possible
memory safety, performance, modern language for low-level applications, ...

I wanted to contribute something back to the community,
i.e, **not** throw away the Python code 1 week after the seminar

The original plan

1. Select the most popular optimization package
2. Implement a simple interior-point method
3. Compare it with Ipopt
4. Profit!



argmin v0.11.0

Numerical optimization in pure Rust

[Homepage](#) [Documentation](#) [Repository](#)

📥 All-Time: 3,049,452

📥 Recent: 531,942

🔄 Updated: 7 months ago

Pure Rust 🦀

We aim at providing a wide range of solvers implemented in pure Rust.

Website ⁷

Consistent interface 🐦

Easily plug your optimization problem in different solvers thanks to a consistent interface.

Type agnostic 🔧

Represent your variables, gradients, Jacobians and Hessians using your own types.

Various math backends 🇧🇷

Use Vecs, ndarray, or nalgebra for behind-the-scenes math. Or implement your own backend.

Observers 📺

Observe the progress of your optimization by logging to screen or file or implement your own observer.

Checkpointing 🚩

Mitigate the negative effects of crashes in unstable computing environments by saving regular checkpoints.

Extensible 🦋

Most features can be exchanged for user supplied implementations thanks to Rusts powerful generics and traits.

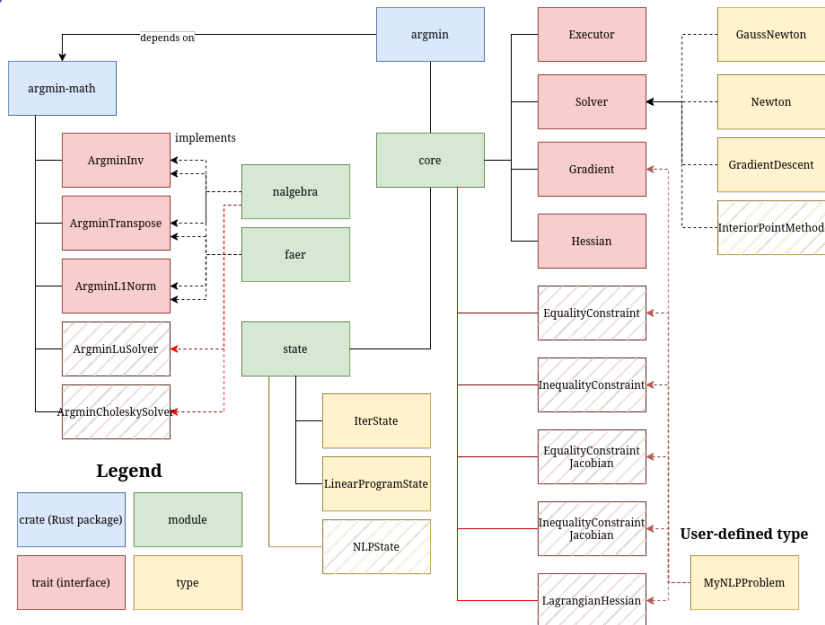
Development framework 🛠️

Use argmins tools and ecosystem to implement your own solver – with all the features mentioned above.

discarded several libraries: dealt only with LP (good_lp), used novel algorithms (clarabel), or were too small (interiors)

⁷<https://argmin-rs.org/>

argmin: Architecture



Math backends

The math backend is implemented in a separate crate `argmin-math`

It is trait-based: interfaces define the methods required from the linear algebra backend

- ▶ `nalgebra` ⁸

- ▶ `faer` ⁹

Following the existing architecture, every matrix operation should be generic over the backend.

But for this toy implementation it would be overkill ...

I decided to take a shortcut and implement a solver tightly coupled to `nalgebra`.

⁸<https://nalgebra.rs/>

⁹<https://faer.veganb.tw/>

argmin: Code example

```
1  impl<O, P, G, H, F> Solver<O, IterState<P, G, (), H, (), F>> for Newton<F>
2  where
3      O: Gradient<Param = P, Gradient = G> + Hessian<Param = P, Hessian = H>,
4      P: Clone + ArgminScaledSub<P, F, P>,
5      H: ArgminInv<H> + ArgminDot<G, P>,
6      F: ArgminFloat,
7  {
8      fn next_iter(
9          &mut self,
10         problem: &mut Problem<O>,
11         mut state: IterState<P, G, (), H, (), F>,
12     ) -> Result<(IterState<P, G, (), H, (), F>, Option<KV>), Error> {
13         let param = state.take_param().ok_or_else(argmin_error_closure!(
14             NotInitialized,
15             concat!(
16                 "`Newton` requires an initial parameter vector. ",
17                 "Please provide an initial guess via `Executor`s `configure` method."
18             ))?;
19         let grad = problem.gradient(&param)?;
20         let hessian = problem.hessian(&param)?;
21         let new_param = param.scaled_sub(&self.gamma, &hessian.inv()?.dot(&grad));
22         Ok((state.param(new_param), None))
23     }
24 }
25 }
```

Notice: The matrix is being inverted at every step (inefficient!). No linear solver $Ax = b$ in the codebase.

Add a linear solver

Add a trait:

```
1  pub trait ArgminLuSolve<T> {  
2      /// Solve linear system using LU decomposition.  
3      fn lu_solve(&self, rhs: &T) -> Result<T, Error>;  
4  }
```

And the corresponding implementation:

```
1  impl<N, D, C, S> ArgminLuSolve<OMatrix<N, D, C>> for SquareMatrix<N, D, S>  
2  where  
3      N: ComplexField,  
4      D: Dim + DimMin<D, Output = D>,  
5      C: Dim,  
6      S: Storage<N, D, D>,  
7      DefaultAllocator: Allocator<N, D, D> + Allocator<N, D, C> + Allocator<N, D>,  
8  {  
9      #[inline]  
10     fn lu_solve(&self, rhs: &OMatrix<N, D, C>) -> Result<OMatrix<N, D, C>, Error> {  
11         let lu = LU::new(self.clone_owned());  
12         lu.solve(rhs).ok_or_else(|| LuSolveError {}.into())  
13     }  
14 }
```

No constraints?

```
1  /// Defines equality constraints for a problem.
2  ///
3  /// It follows the form  $h(x) = (0, \dots, 0)$ , where  $h$  is a vector field.
4  pub trait EqualityConstraint {
5      /// Type of the parameter vector
6      type Param;
7      /// Type of the return value of the equality constraints
8      type Output;
9
10     /// Compute all equality constraints
11     fn equality_constraint(&self, param: &Self::Param) -> Result<Self::Output, Error>;
12 }
```

And their derivatives ...

```
1  /// Defines the Jacobian of the vector field representing the equality constraints for a
2  ↳ problem.
3  pub trait EqualityConstraintJacobian {
4      /// Type of the parameter vector
5      type Param;
6      /// Type of the Jacobian
7      type Jacobian;
8
9      /// Compute the Jacobian of the equality constraints
10     fn equality_constraint_jacobian(&self, param: &Self::Param) -> Result<Self::Jacobian,
11     ↳ Error>;
12
13     bulk!(equality_constraint_jacobian, Self::Param, Self::Jacobian);
14 }
```


And the Hessian of the Lagrangian ...

The existing Hessian is not enough. We need Lagrange multipliers as well ...

```
1  pub trait LagrangianHessian {
2      /// Type of the parameter vector
3      type Param;
4      /// Type of the multipliers associated with equality constraints
5      type MultipliersEq;
6      /// Type of the multipliers associated with inequality constraints
7      type MultipliersIneq;
8      /// Type of the Hessian
9      type Hessian;
10
11     /// Compute the Hessian of the Lagrangian
12     fn lagrangian_hessian(
13         &self,
14         param: &Self::Param,
15         multipliers_eq: &Self::MultipliersEq,
16         multipliers_ineq: &Self::MultipliersIneq,
17     ) -> Result<Self::Hessian, Error>;
18
19     bulk!(
20         lagrangian_hessian,
21         Self::Param,
22         Self::MultipliersEq,
23         Self::MultipliersIneq,
24         Self::Hessian
25     );
26 }
```

And iteration state for NLP ...

Lots of new things to keep track of!

- ▶ `slacks`
- ▶ `best_slacks`
- ▶ `lagrangian_hessian`
- ▶ `equality_constraints`
- ▶ `inequality_constraints`
- ▶ `equality_constraint_jacobian`
- ▶ `inequality_constraint_jacobian`
- ▶ `mu`
- ▶ `lambda_eq`
- ▶ `lambda_ineq`
- ▶ `alpha_primal`
- ▶ `alpha_dual`
- ▶ `inf_pr`
- ▶ `inf_du`
- ▶ `compl_inf`

The resulting `NonLinearProgramState` is generic over 11 types!

Finally! IPM

`nalgebra_backend.rs`: All the linear-algebra-specific logic for `nalgebra`, the only supported backend

`linesearch.rs`: The custom backtracking linear search. The existing implementation could not be reused

`iteration.rs`: A single iteration of the interior-point method

`interiorpointmethod.rs`: The solver. The high-level management of the IPM algorithm.

Results: TODO

This is where the results of the implementation would go.

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riopt: A Rust re-write from February 2026

Coded in $\sim 60/70$ hs during the course of 6 weeks heavily relying on Claude Code ¹⁰.

Opus 4.6 with a 1M context window. Several agents running in parallel.

License? EPL, same as lpopt. Author considers it a derivative work.

Linear algebra solver?

- ▶ First relied on a reimplement of MUMPS ¹¹
- ▶ Since April 26, it relies on a separate AI-coded solver by the same author ¹²

¹⁰<https://github.com/jkitchin/riopt>

¹¹<https://mumps-solver.org/>

¹²<https://github.com/jkitchin/feral>

riopt: Licensing

Author's impression:

For small problems there are compelling reasons to be excited about riopt. For larger problems, there is still a lot of work to do to make it competitive.

Interesting insights into thought process documented in the repo.

It is not a "clean room" development.

Caution!

I have not read all the generated code (I don't even really know how to write Rust, but it is readable). It was generated faster than I could read or understand.

riopt: Benchmarks used

Benchmarks and testing are key to make sure that it actually works!

- ▶ Hock-Schittkowski¹³ test suite (120 problems / 306 in total)
- ▶ CUTEst 745 problems¹⁴
- ▶ Electrical grid optimization problems (4 problems)¹⁵
- ▶ Gas pipeline problems (4 problems)
- ▶ Electrolyte thermodynamics problems (13 problems)
- ▶ Parameter estimation for a CHO (Chinese Hamster Ovary) cell kinetic model (1 problem)
- ▶ Water distribution network problems (6 problems)

We will see how much of this holds up to scrutiny....

¹³Schittkowski [2012](#); Schittkowski [2008](#).

¹⁴Gould, Orban, and Toint [2015](#).

¹⁵Zimmerman, Murillo-Sánchez, and Thomas [2010](#).

It's just replicating the results, right?

`make benchmark` does not work anymore, targets in Makefile not found #37

Open



hlisdero opened 5 days ago

Last edited by hlisdero ...

I would like to run the full benchmark suite with the following command to verify the results obtained.

```
make benchmark
```



Unfortunately, none of the benchmark-related targets are present in the Makefile anymore. It also contains some outdated targets like `cutest-install` (script `install_cutest.sh` not present in the repo).

I also tried running a cargo command like:

```
cargo run --bin cutest_suite --features cutest,ipopt-native --release
```



But this lead to the issue of not having CUTEst installed which lead me in turn to the missing bash script `install_cutest.sh`.

How can I work around this?



jkitchin closed this as completed in [eb1979e](#) 5 days ago



jkitchin 5 days ago

Last edited by jkitchin ... Owner ...

Thanks for the report. The root `Makefile` does still have the `benchmark` / `cutest` targets, but they are chime that delegate

Assignees

No one assigne

Labels

No labels

Projects

No projects

Milestone

No milestone

Relationships

None yet

Development

No branches or

Participants



How it got fixed

After ...

- ▶ a few hallucinated links,
- ▶ understanding how to install CUTEst locally and compiling the Fortran code,
- ▶ pointing out missing scripts,
- ▶ discovering a Linux-specific issue when linking the ipopt library,
- ▶ testing the automated CUTEst installation flow from scratch,
- ▶ finding a collision in the name of two Fortran functions,
- ▶ testing every Makefile target,
- ▶ spotting the missing Hoch-Schittkowski test suite (removed)...

CUTEst benchmark summary

Metric	riopt	lpopt
Optimal	562/745 (75.4%)	574/745 (77.0%)
Acceptable	16	3
Total solved (Optimal + Acceptable)	578 (77.6%)	577 (77.4%)
Solved exclusively	21	20
Both solved	557	
Matching objectives (<0.01%)	537/557	
Acceptable at worse local min	4	

Note: rlopt uses fallback strategies (L-BFGS Hessian, Augmented Lagrangian, SQP, slack reformulation) that lpopt does not have, which accounts for much of the Acceptable count difference.

Acceptable? This was inherited from lpopt's own convergence framework: a solve is *Optimal* if it meets tight tolerances, and *Acceptable* if it meets relaxed tolerances.

CUTEst performance

Metric	riopt	lpopt
Median time	289 μ s	2.9 ms
Total time	29.76 s	27.98 s
Mean iterations	38.6	37.1
Median iterations	13	12

- ▶ **Geometric mean speedup:** $7.5\times$
- ▶ **Median speedup:** $10.5\times$
- ▶ ripoft faster: 478/541 (88%)
- ▶ ripoft $10\times+$ faster: 288/541
- ▶ lpopt faster: 63/541

Comparison with committed report: The results in the repo look slightly better. The difference could be due to the different hardware, a regression after new commits, and partly AI hallucination.

Other test suites

Suite	N	riopt	lpopt	r-only	i-only	Both	Match
CUTEst	727	561 (77.2%)	561 (77.2%)	20	20	541	522/541
Electrolyte	13	13 (100.0%)	12 (92.3%)	1	0	12	11/12
Grid	4	4 (100.0%)	4 (100.0%)	0	0	4	4/4
CHO	1	0 (0.0%)	0 (0.0%)	0	0	0	0/1

Remarks:

- ▶ r-only: number of problems which only rlopt could solve
- ▶ i-only: number of problems which only lpopt could solve
- ▶ CHO problem also not solved in report in repo. Probably a bug?

Electrolyte and Grid performance

Electrolyte suite (12 problems) ^a

Metric	riopt	lpopt
Median time	121 μ s	1.5 ms
Total time	30.1 ms	82.0 ms
Mean iterations	203.1	27.9
Median iterations	7	7

- ▶ **Geometric mean speedup:** 11.1 $\times$
- ▶ **Median speedup:** 14.8 $\times$
- ▶ riopt faster: 11/12 (92%)
- ▶ riopt 10 $\times$ + faster: 11/12
- ▶ lpopt faster: 1/12

Grid suite (4 problems)

Metric	riopt	lpopt
Median time	19.0 ms	16.2 ms
Total time	114.7 ms	157.4 ms
Mean iterations	15.8	12.5
Median iterations	19	14

- ▶ **Geometric mean speedup:** 1.8 $\times$
- ▶ **Median speedup:** 1.8 $\times$
- ▶ riopt faster: 3/4 (75%)
- ▶ riopt 10 $\times$ + faster: 0/4
- ▶ lpopt faster: 1/4

^a12 instead of 13, unclear why

CUTEst Suite: failure modes

Failure Mode	riopt	lpopt
ErrorInStepComputation	0	2
EvaluationError	3	0
Infeasible	0	10
InvalidNumberDetected	0	1
lpoptStatus(-10)	0	123
lpoptStatus(4)	0	3
LocalInfeasibility	27	0
MaxIterations	22	12
MaxTimeExceeded	15	0
NumericalError	11	0
RestorationFailed	70	4
StopAtTinyStep	1	0
Timeout	17	11

lpoptStatus(-10) = not enough degrees of freedom (typically overconstrained model). lpoptStatus(4) = diverging iterates. InvalidNumberDetected = NaN/Inf in functions or derivatives. RestorationFailed = feasibility restoration failed.

Large scale problems: Solver comparison

Problem	n	m	riopt	iters	time	lpopt	iters	time
Rosenbrock 500	500	0	Optimal	751	0.127s	Optimal	749	0.280s
SparseQP 1K	500	500	Optimal	6	0.007s	Optimal	6	0.006s
Bratu 1K	1,000	998	Optimal	2	0.004s	Optimal	2	0.003s
OptControl 2.5K	2,499	1,250	Optimal	1	0.017s	Optimal	1	0.004s
Rosenbrock 5K	5,000	0	MaxIterations	2999	5.340s	Failed	3000	7.093s
Poisson 2.5K	5,000	2,500	Optimal	1	0.104s	Optimal	1	0.018s
Bratu 10K	10,000	9,998	Optimal	1	0.039s	Optimal	2	0.023s
OptControl 20K	19,999	10,000	Optimal	1	0.167s	Optimal	1	0.034s
Poisson 50K	49,928	24,964	Optimal	1	2.416s	Optimal	1	0.248s
SparseQP 100K	50,000	50,000	Optimal	6	1.192s	Optimal	6	0.638s

- ▶ rlopt: **9/10 solved** in 9.4s total
- ▶ lpopt: **9/10 solved** in 8.3s total
- ▶ Speedups (time lpopt/time rlopt):
2.2x, 0.8x, 0.7x, 0.3x, 1.3x, 0.2x, 0.6x, 0.2x, 0.1x, 0.5x

Bottom line: rlopt is slower with larger problems due to less performant linear solver

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Conclusion

Ipopt's performance can still be improved.

Different methods complement each other: By combining them, edge cases can be solved more efficiently.

Implementation of a state-of-the-art optimization package may be in the reach of modern AI agentic coding tools.

Questions?

Links

Presentation:

https://github.com/hlisdero/JMCS_Seminar_Applied_Optimization

Modified argmin:

<https://github.com/hlisdero/argmin>

riopt:

<https://github.com/jkitchin/riopt>

ipopt-rs:

<https://github.com/elrnv/ipopt-rs>

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